

ASSIGNMENTS – OOADI

LECTURE 5

1. Trains can be characterized in many ways and includes locomotives and wagons of all sorts (goods, passengers, luxury wagons, armored wagons, dangerous goods etc.). Sketch first a class diagram for defining Trains.
2. One possible superclass to a Train class could be Vehicle. Define an interface that properly describes what vehicles can do. Let the Train implement the Vehicle interface.
3. Consider that some of your trains may be electric powered, and some may be diesel (and others). Define an electric Train interface that captures the electric type, e.g. it requires charging, one needs to check status of the battery etc. and implement on a few selected Train classes (or subclasses) that interface.
4. The electric Train may need to be initiated differently as the normal version of the Train, e.g. starting charge state of the battery which is not present in the diesel version of the train. Use polymorphism to ensure your Train class can support the electric Train version in a generic fashion.
5. Will there be a difference if the top-level Vehicle class is created as an abstract class? Is this a reasonable choice? Why/why not?
6. Create other vehicle types – just a few examples, e.g. bicycles from previous assignments. Which of the operations from previous assignments can you easily perform?